Floquet theory aims at finding solutions to the time-dependent Schrödinger equation with a time-periodic Hamiltonian:

$$i\hbar \frac{d}{d\tau} |\psi(\tau)\rangle = H(\tau)|\psi(\tau)\rangle,$$

$$H(\tau) = H(\tau+T)$$

$$\mathcal{U}(\tau_1, \tau_0) = \mathcal{T} \exp\left[-\frac{i}{\hbar}\int_{\tau_0}^{\tau_1} H(\tau)\, d\tau\right], \quad H(\tau) = H(\tau+T)$$

$$* \ \mathcal{U}(\tau_1+\tau_2, \tau_0) = \mathcal{U}(\tau_1+\tau_2, \tau_1)\,\mathcal{U}(\tau_1, \tau_0)$$

$$\mathcal{U}(\tau+T, \tau_0) = \mathcal{U}(\tau+T, \tau_0+T)\,\mathcal{U}(\tau_0+T, \tau_0) = \mathcal{U}(\tau, \tau_0)\,\mathcal{U}(\tau_0+T, \tau_0)$$

$$\mathcal{U}(\tau_0+T, \tau_0) = e^{-iH_F T/\hbar}, \quad H_F: \text{time-independent, Hermitian operator}$$

Fast-motion operator $e^{-i\hat{K}_F(\tau)} \equiv \mathcal{U}(\tau, \tau_0)\, e^{iH_F \times (\tau-\tau_0)/\hbar}$

↳ $\hat{K}_F(\tau)$ is called the stroboscopic kick operator

$$e^{-i\hat{K}_F(\tau+T)} = \mathcal{U}(\tau+T, \tau_0)\, e^{iH_F \times (\tau+T-\tau_0)/\hbar}$$

$$= \mathcal{U}(\tau, \tau_0)\, e^{-iH_F T/\hbar}\, e^{iH_F \times (\tau+T-\tau_0)/\hbar}$$

$$= \mathcal{U}(\tau, \tau_0)\, e^{iH_F \times (\tau-\tau_0)/\hbar} = e^{-i\hat{K}_F(\tau)}$$

$$\begin{cases} \mathcal{U}(\tau, \tau_0) = e^{-i\hat{K}_F(\tau)}\, e^{-iH_F \times (\tau-\tau_0)/\hbar}, \text{ with} \\ e^{-i\hat{K}_F(\tau+T)} = e^{-i\hat{K}_F(\tau)} \end{cases}$$

The Floquet gauge τ_0

All choices of τ_0 can be related via a unitary transformation

$$H_F[\tau_0'] = \mathcal{U}(\tau_0', \tau_0)\, H_F[\tau_0]\, \mathcal{U}(\tau_0, \tau_0')$$

$$\begin{cases} \mathcal{U}(\tau_0+T, \tau_0) = e^{-iH_F[\tau_0]T/\hbar} \\ e^{-i\hat{K}_F[\tau_0](\tau)} = \mathcal{U}(\tau, \tau_0)\, e^{iH_F[\tau_0] \times (\tau-\tau_0)/\hbar} \end{cases}$$

$$\Rightarrow \begin{cases} \mathcal{U}(\tau, \tau_0) = e^{-i\hat{K}_F[\tau_0](\tau)}\, e^{-iH_F[\tau_0] \times (\tau-\tau_0)/\hbar} \\ \hat{K}_F[\tau_0](\tau+T) = \hat{K}_F[\tau_0](\tau) \end{cases}$$

Floquet states and quasienergies

$$U(\tau_0+T, \tau_0) = e^{-iH_F T/\hbar}$$ unitary $\Rightarrow$ eigenvalues $\in \mathbb{C}$

$$\{e^{-i\epsilon_n T/\hbar}\}$$

$\{\epsilon_n$, mod $\hbar\omega\}$: quasienergies

$$H_F|n\rangle = \epsilon_n|n\rangle$$

$$|\psi(\tau_0)\rangle = \sum_n |n\rangle\langle n|\psi(\tau_0)\rangle = \sum_n a_n|n\rangle$$

$$\begin{aligned}
|\psi(\tau)\rangle &= U(\tau, \tau_0)|\psi(\tau_0)\rangle \\
&= e^{-i\hat{K}_F(\tau)} e^{-iH_F\times(\tau-\tau_0)/\hbar} \sum_n a_n|n\rangle \\
&= \sum_n a_n e^{-i\hat{K}_F(\tau)} e^{-i\epsilon_n(\tau-\tau_0)/\hbar}|n\rangle \\
&= \sum_n a_n e^{-i\epsilon_n(\tau-\tau_0)/\hbar}|u_n(\tau)\rangle
\end{aligned}$$

where $\begin{cases} |u_n(\tau)\rangle \equiv e^{-i\hat{K}_F(\tau)}|n\rangle & \text{"Floquet modes"} \\ \text{with} \quad |u_n(\tau)\rangle = |u_n(\tau+T)\rangle \end{cases}$

Numerically calculating the Floquet Problem

△ Exact time-evolution of $U(\tau_0+T, \tau_0)$

* $\mathbb{H}$ is not too big.

① Expand the initial state in a certain basis

$$|\psi(\tau_0)\rangle = \sum_n C_n(\tau_0)|n\rangle$$

② Evolve $\psi(\tau_0)$ over one period according to the Schrödinger equation

$$i\hbar \frac{d}{d\tau} C_n(\tau) = \sum_m H_{nm}(\tau) C_m(\tau)$$

for each $C_n(\tau_0)$ from τ_0 to τ_0+T

③ The time-evolved vectors $\{C_n(\tau_0+T)\}$ directly give the columns of

$U(\tau_0+T, \tau_0)$ which, upon diagonalization, give the Floquet multipliers $\{ e^{-i\epsilon_n T/\hbar} \}$.

△ Trotter decomposition of $U(\tau_0+T, \tau_0)$

* $|H|$ is too large.

Divide the timespan $[\tau_0, \tau_0+T]$ into N segments $\begin{cases} \tau_i = \tau_0 + i\Delta\tau \\ \Delta\tau = T/N \end{cases}$

$$U(\tau_0+T, \tau_0) = \mathcal{T} \exp\left[-\frac{i}{\hbar} \int_{\tau_0}^{\tau_0+T} H(\tau)\, d\tau \right]$$

$$\simeq \exp\left[-\frac{i}{\hbar} \sum_{i=0}^{N-1} H(\tau_i)\, \Delta\tau \right]$$

$$= \prod_{i=0}^{N-1} \exp\left[-\frac{i}{\hbar} H(\tau_i)\, \Delta\tau \right] + \mathcal{O}(\Delta\tau^2)$$

$H(t) = H(t+T)$

Hermitian.

$$U(T) = \mathcal{T} \exp\left[-i\int_0^T H(s)\,ds\right] = \exp(-i H_F T)$$

$$U(T)|\phi_\alpha\rangle = e^{-i\epsilon_\alpha T}|\phi_\alpha\rangle$$

$$|\phi_\alpha(T)\rangle = |\phi_\alpha(0)\rangle \quad , \quad |\psi_\alpha(t)\rangle = e^{-i\epsilon_\alpha t}|\phi_\alpha(t)\rangle$$

$$(\,|\psi_\alpha(0)\rangle = |\phi_\alpha\rangle$$

ϵ_α : real, $\{|\phi_\alpha(0)\rangle\}$ is an orthonormal basis for $\mathcal{H}$.

$$e^{-itH_F} = \sum_\alpha e^{-it\epsilon_\alpha}|\phi_\alpha(0)\rangle\langle\phi_\alpha(0)|$$

$H(t) = H(t+T)$

$|\psi_n(t)\rangle = |u_n(t)\rangle e^{-i\varepsilon_n t/\hbar}$, $|u_n(t)\rangle = |u_n(t+T)\rangle$

$\hat{U}(t_0+T, t_0)\,|\psi_n(t_0)\rangle = e^{-i\varepsilon_n T/\hbar}\,|\psi_n(t_0)\rangle$

$\hat{U}(t_2, t_1) = \sum_n e^{-i\varepsilon_n(t_2-t_1)/\hbar}\,|u_n(t_2)\rangle\langle u_n(t_1)|$

$|\psi(t)\rangle = \sum_n c_n e^{-i\varepsilon_n(t-t_0)/\hbar}\,|u_n(t)\rangle$

$\hookrightarrow c_n = \langle u_n(t_0)|\psi(t_0)\rangle$

$\hat{H}^F_{t_0}$: $e^{-i\hat{H}^F_{t_0}T/\hbar} = \hat{U}(t_0+T, t_0)$

$\hat{H}^F_{t_0} = \sum_n \varepsilon_n |u_n(t_0)\rangle\langle u_n(t_0)|$ $\qquad \hat{H}^F_{t_0+T} = \hat{H}^F_{t_0}$

two-point micromotion operator $\hat{U}_F(t_2, t_1) = \sum_n |u_n(t_2)\rangle\langle u_n(t_1)|$

$|u_n(t_2)\rangle = \hat{U}_F(t_2, t_1)\,|u_n(t_1)\rangle$

starting from the known solution given by $|u_n(t)\rangle$ and ε_n, one can label all possible choices for the quasienergy by introducing the integer index m,

$$\varepsilon_{nm} = \varepsilon_n + m\hbar\omega$$

$$|u_{nm}(t)\rangle = |u_n(t)\rangle e^{im\omega t}$$

$\Rightarrow |\psi_n(t)\rangle = |u_n(t)\rangle e^{-i\varepsilon_n t/\hbar} = |u_{nm}(t)\rangle e^{-i\varepsilon_{nm} t/\hbar}$

$\Rightarrow \left(\hat{H}(t) - i\hbar\frac{d}{dt}\right)|u_{nm}(t)\rangle = \varepsilon_{nm}|u_{nm}(t)\rangle$

extended Hilbert space $\mathcal{F} = \mathbb{H} \otimes \mathcal{L}^2_T$

$\langle\langle u|v\rangle\rangle_{\mathcal{F}} = \frac{1}{T}\int_0^T dt\, \langle u(t)|v(t)\rangle$

$\bar{Q}$: operator in $\mathcal{F}$

$$|\alpha\, m(t)\rangle = |\alpha\rangle e^{im\omega t}$$

$$\langle\langle \alpha' m' | \bar{Q} | \alpha m \rangle\rangle = \frac{1}{T}\int_0^T dt\, e^{-im'\omega t} \langle \alpha' | \hat{H}(t) - i\hbar \frac{d}{dt} | \alpha \rangle e^{im\omega t}$$

$$= \langle \alpha' | \hat{H}_{m'-m} | \alpha \rangle + \delta_{m'm}\delta_{\alpha'\alpha}\, m\hbar\omega$$

$$\hat{H}_m = \frac{1}{T}\int_0^T dt\, e^{-im\omega t}\hat{H}(t) = \hat{H}^{+}_{-m}$$

i.e. $\hat{H}(t) = \sum_{m=-\infty}^{\infty} e^{im\omega t}\hat{H}_m$

$$\bar{Q} \to \bar{Q}' = \bar{U}^{+}\bar{Q}\bar{U}$$
$$|u\rangle\rangle \to |u'\rangle\rangle = \bar{U}^{+}|u\rangle\rangle$$

$$\Longleftrightarrow \begin{cases} \hat{H}(t) = \hat{H}'(t) = \hat{U}^{+}(t)\hat{H}(t)\hat{U}(t) - i\hbar\hat{U}^{+}(t)\, d_t\, \hat{U}(t) \\ |\psi(t)\rangle \longrightarrow |\psi'(t)\rangle = \hat{U}^{+}(t)\, |\psi(t)\rangle \end{cases}$$